

## Divide quantities in a given ratio



- 1 James and Emma scored goals in their netball game in the ratio 4 : 3.
  - a Find the fraction of the goals scored by James.
  - b If they scored a total of 56 goals, how many goals did James score?
  - c How many goals did Emma score?
  
- 2 Harris and Bella divided lollies between them in the ratio 5 : 4.
  - a Find the fraction of lollies that Bella gets.
  - b If there are 27 lollies, how many does Bella get?
  - c How many lollies does Harris get?
  
- 3 56 building blocks are shared between Mohamad and Isabelle in the ratio 2 : 5.
  - a Find the fraction of the blocks Mohamad receive.
  - b How many blocks does Mohamad receive?
  - c How many blocks does Isabelle receive?
  
- 4 Ben always buys melons and bananas in the ratio 7 : 3. If he buys 30 pieces of fruit in total, how many melons did he buy?
  
- 5 The ratio of adults to children on a train is 2 : 7. The train is carrying 360 passengers.
  - a How many adults are on the train?
  - b How many children are on the train?
  
- 6 James and Sophia share \$880 in the ratio 2 : 9.
  - a How much money does James get?
  - b How much money does Sophia get?
  
- 7 Avril is making a scarf that uses two materials. She wants to use the materials in the ratio of  $\frac{11}{15}$  metres of red for every  $\frac{3}{5}$  metres of green.
  - a Write the simplified ratio of red material to green material.
  - b If she needs 33 metres of red material, how many metres of green material will she need?



- 8 Neil is mixing paint to get a specific shade of orange. He wants a mix of red and yellow paint in the ratio of  $3\frac{3}{4} : 6\frac{2}{3}$ .
- Simplify the ratio.
  - How many litres of red paint will Neil need if he wants to make 75 litres of orange paint?
- 9 Kathleen and Luke invest money in a business in the ratio 3 : 10.
- What fraction did Kathleen contribute?
  - If Kathleen has invested \$2100, how much did Luke invest?
- 10 At a sporting event, the ratio of security personnel to spectators is 1 : 180. If 34 200 spectators attend the event, how many security personnel will be required?
- 11 Two kinds of pine trees, type A and type B, are planted in rows. In each row the ratio of type A to type B is 9 : 8 Altogether, 180 type A trees are planted.
- How many type B trees are planted?
- 12 Xavier and Ursula split the cost of the groceries in the ratio 0.5 : 1.5. Altogether the groceries cost \$140.
- Simplify the ratio.
  - How much did Xavier pay for the groceries?
- 13 A car and a truck are driving down the highway. Their relative speeds are in the ratio 3.1 : 2.9. If the car is travelling at 93 km/hr, how fast is the truck travelling?
- 14 In a zoo the ratio of elephants to lions is 7 : 4.
- Find the ratio of elephants to lions to the total number of elephants and lions.
  - Complete the following table of equivalent ratios:

| Elephants | to | Lions | to | Total |
|-----------|----|-------|----|-------|
| 7         | :  | 4     | :  |       |
| 14        | :  |       | :  |       |
|           | :  |       | :  | 110   |

- If there are 66 elephants and lions altogether, how many lions are there?

**15** The ratio of red, green and black coloured  on a dart board is 12 : 7 : 31. The area of the dart board is  $150 \text{ cm}^2$ .

**a** Find the number of ratio parts corresponding to the following:

- i** Red area      **ii** Green area      **iii** Black area      **iv** Total area

**b** Find the size of a single ratio part.

**c** Find the area of the following:

- i** Red area      **ii** Green area      **iii** Black area

**16** Vanessa competed in a marathon and found that her times for the swimming, cycling and running parts were in the ratio 3 : 1 : 2. If the quickest part took 10 minutes, how many minutes did she take to complete the whole marathon?

**17** The ratio of students who play badminton, football and softball is 2 : 5 : 9.

**a** If there are 72 students who play softball, how many students play badminton?

**b** Find the total number of students who play one of these sports.

**18** The perimeter of a rectangle is 42 cm and the ratio of its length to its width is 4 : 3.

**a** How many parts are in the ratio?

**b** Find the length of the rectangle.

**c** Find the width of the rectangle.

---

## Increase and decrease in a given ratio (Extended)

**19** Increase the following quantities in the ratio specified:

**a** \$400 in the ratio 3 : 1.

**b** \$780 in the ratio 5 : 2.

**c** 60 kg in the ratio 4 : 3.

**d** 800 g in the ratio 3 : 2.

**e** 1.5 L in the ratio 2 : 1.

**f** 1200 mL in the ratio 6 : 5.

**g** 120 cm in the ratio 10 : 2.

**h** 5.2 m in the ratio 100 : 5.

**20** Decrease the following quantities in the ratio specified:

**a** \$500 in the ratio 1 : 5.

**b** \$820 in the ratio 3 : 4.

**c** 50 kg in the ratio 2 : 5.

**d** 150 g in the ratio 1 : 2.

**e** 2.4 L in the ratio 5 : 6.

**f** 900 mL in the ratio 4 : 5.

**g** 20 cm in the ratio 8 : 10.

**h** 2.4 m in the ratio 5 : 20.



**21** A clothing store is having a sale and is decreasing the price of all their items in the ratio 3 : 4. Find the new sale price of the following items:

- a** A jumper originally priced at \$50.
- b** A dress originally priced at \$120.
- c** A pair of jeans originally priced at \$80.
- d** A skirt originally priced at \$75.

**22** At the beginning of the year, an office supplies store increases the prices of its items in the ratio of 7 : 6. Find the new increased prices of the following items:

- a** A pencil case originally priced at \$12.
- b** A desk originally priced at \$180.
- c** A pair of jeans originally priced at \$54.
- d** A set of shelves originally priced at \$150.